

Sensor Modality Selection (Sc-02) Data Imputed – Range

Scenario: Lane Change in front of the Ego car on US101 Highway - Data Imputed																					
Trajectory Prediction Model		CS-LSTM										STDAN									
Manouver		Right Lane change					Left Lane change					Right Lane change					Left Lane change				
Sensor Parameter		RMSE (m) at 'Range'					RMSE (m) at 'Range'					RMSE (m) at 'Range'					RMSE (m) at 'Range'				
Sensor Modality	Prediction Horizon (s)	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m
Radar	1	0.33	0.31	0.32	0.3	0.31	0.3	0.31	0.32	0.3	0.32	0.35	0.35	0.35	0.35	0.35	0.38	0.38	0.39	0.38	0.38
	2	0.76	0.7	0.75	0.68	0.7	0.67	0.68	0.73	0.66	0.69	1.14	1.14	1.13	1.15	1.14	1.02	1.02	1.03	1.02	1.02
	3	1.32	1.26	1.31	1.21	1.25	1.22	1.23	1.28	1.22	1.24	2.33	2.33	2.31	2.34	2.32	1.82	1.81	1.83	1.82	1.81
	4	1.72	1.57	1.62	1.55	1.6	1.6	1.63	1.71	1.59	1.63	3.35	3.35	3.32	3.36	3.34	2.52	2.52	2.54	2.54	2.52
	5	2.45	2.27	2.26	2.22	2.26	2.31	2.37	2.48	2.31	2.34	3.53	3.53	3.49	3.56	3.53	3.02	3.02	3.02	3.01	3.01
Camera	1	0.53	0.51	0.53	0.51	0.54	0.5	0.52	0.53	0.46	0.55	1.71	1.63	1.53	1.66	1.76	1.7	1.66	1.76	1.59	1.76
	2	1.21	1.15	1.22	1.23	1.21	1.12	1.15	1.13	1.04	1.14	3.26	3.42	3	3.31	3.63	3.36	3.16	3.66	3.27	3.44
	3	1.76	1.72	1.77	1.87	1.81	1.75	1.78	1.73	1.63	1.77	4.86	5.11	4.18	4.73	5.37	5.32	4.92	5.99	5.09	5.38
	4	2.43	2.32	2.38	2.52	2.51	2.48	2.34	2.4	2.22	2.51	7.21	7.78	6.04	6.85	7.84	8.53	7.64	9.46	7.97	8.55
	5	3.4	3.13	3.32	3.48	3.48	3.37	3.22	3.26	2.95	3.42	10.74	11.18	9.11	9.85	11.26	12.28	11.7	13.22	11.67	12.45

Sensor Modality Selection (Sc-02) Data Imputed – HFOV

Scenario: Lane Change in front of the Ego car on US101 Highway - Data Imputed																									
Trajectory Prediction Model		CS-LSTM												STDAN											
Manouver		Right Lane change						Left Lane change						Right Lane change						Left Lane change					
Sensor Parameter		RMSE (m) at 'HFOV'						RMSE (m) at 'HFOV'						RMSE (m) at 'HFOV'						RMSE (m) at 'HFOV'					
Sensor Modality	Prediction Horizon (s)	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg
Radar	1	0.28	0.29	0.28	0.28	0.29	0.28	0.3	0.29	0.29	0.29	0.3	0.3	0.35	0.35	0.35	0.35	0.35	0.35	0.39	0.38	0.39	0.38	0.38	0.39
	2	0.63	0.64	0.62	0.63	0.64	0.63	0.67	0.65	0.65	0.66	0.66	0.66	1.13	1.15	1.13	1.14	1.14	1.13	0.99	0.98	1.02	1.03	1.02	1.02
	3	1.22	1.23	1.22	1.23	1.25	1.23	1.3	1.27	1.28	1.29	1.29	1.28	2.34	2.33	2.32	2.34	2.33	2.32	1.75	1.72	1.81	1.83	1.81	1.81
	4	1.81	1.81	1.79	1.8	1.83	1.8	1.95	1.91	1.91	1.93	1.93	1.93	3.37	3.36	3.34	3.37	3.35	3.33	2.42	2.37	2.53	2.52	2.52	2.52
	5	2.72	2.73	2.71	2.72	2.76	2.72	2.95	2.9	2.91	2.94	2.91	2.95	3.54	3.56	3.52	3.56	3.53	3.5	2.87	2.9	3.02	3	3	3.01
Camera	1	0.73	0.75	0.71	0.71	0.73	0.71	0.7	0.7	0.68	0.69	0.69	0.68	1.87	1.61	1.62	1.6	1.67	1.56	1.83	1.77	1.74	1.73	1.86	1.74
	2	1.42	1.45	1.38	1.37	1.41	1.35	1.37	1.39	1.32	1.36	1.35	1.31	3.84	3.16	3.23	3.29	3.46	3.18	3.68	3.65	3.59	3.61	3.72	3.55
	3	2.18	2.2	2.13	2.13	2.17	2.1	2.17	2.18	2.12	2.17	2.14	2.12	5.87	4.78	4.93	4.92	5.22	4.79	5.67	5.88	5.69	5.87	5.73	5.65
	4	3.16	3.16	3.1	3.09	3.12	3.05	3.16	3.15	3.09	3.15	3.09	3.08	8.59	7.15	7.36	7.34	7.31	7.22	9.06	9.25	8.79	8.76	8.72	8.84
	5	5.04	5.04	4.93	4.95	4.99	4.86	4.93	4.93	4.85	4.87	4.9	4.82	11.97	10.23	10.72	10.55	10.55	10.67	13.49	13.14	12.67	12.2	12.71	12.91